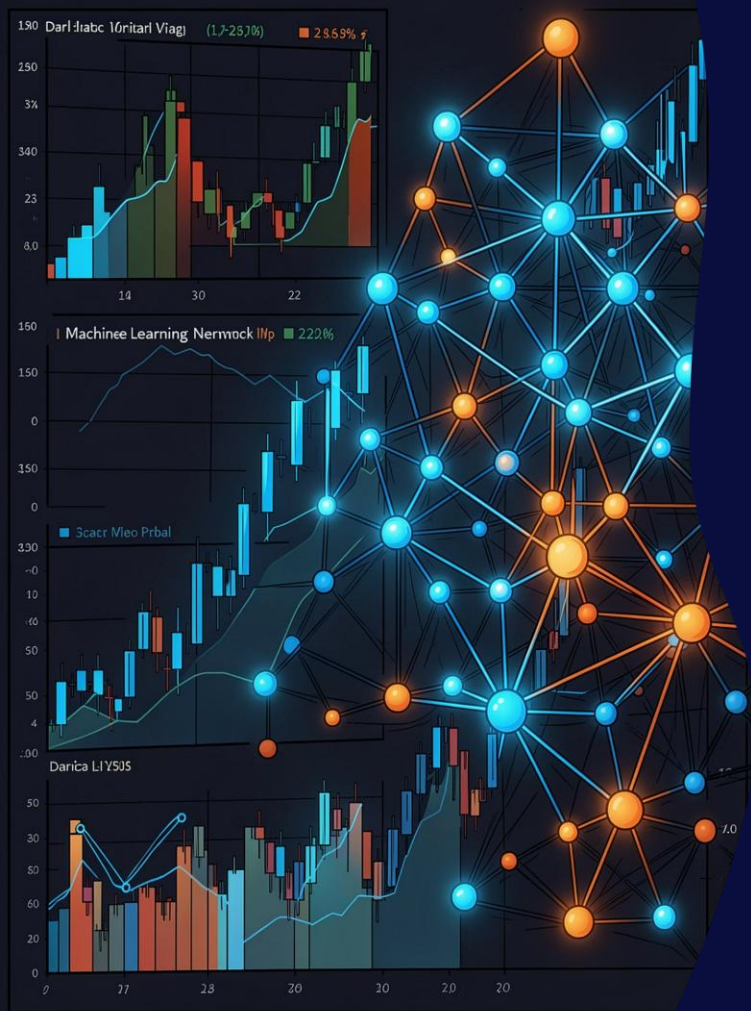




Dynamic Pricing Engine

Using Machine Learning for Revenue Optimization — analyzing real-time and historical data to recommend the optimal price for every product or service.

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INTRODUCTION



Pricing plays a vital role in determining a business's revenue and competitiveness. Traditional fixed pricing strategies often fail to adapt to changing market conditions such as customer demand, competitor pricing, inventory levels, and market trends. The Dynamic Pricing Engine – ML-Based Real-Time Price Optimization uses Machine Learning algorithms such as Random Forest and XGBoost to analyze historical and real-time data and recommend optimal prices automatically. By considering factors like demand, competitor prices, inventory availability, and customer behavior, the system helps businesses maximize revenue, improve pricing efficiency, and make intelligent data-driven decisions through a dynamic pricing dashboard.

Problem Statement

Businesses often use fixed pricing strategies that fail to adapt to changing market conditions. This can lead to lost revenue, excess inventory, or reduced competitiveness.

Lost Revenue

Static prices fail to capture maximum value when market conditions shift.

Excess Inventory

Without demand-aware pricing, stock sits unsold and capital is tied up.

Reduced Competitiveness

Fixed strategies cannot respond to competitor price movements in real time.

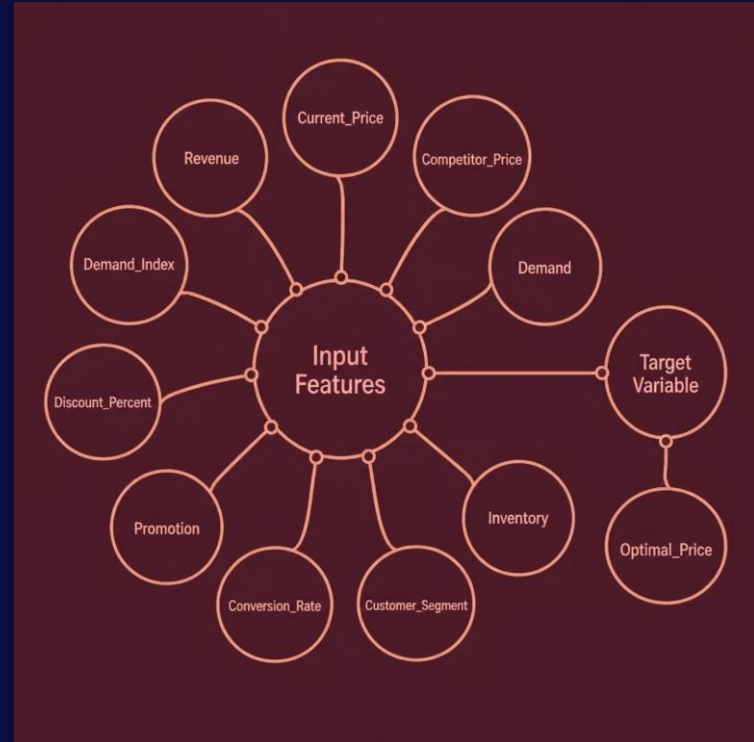


The Dataset

The project uses a **Dynamic Pricing Dataset** (Kaggle) containing business and market-related attributes required for price optimization.

Dataset Features

- Date, Hour, Day_of_Week
- Product_ID, Product
- Current_Price, Competitor_Price
- Demand, Inventory
- Customer_Segment, Conversion_Rate
- Promotion, Discount_Percent
- Demand_Index, Revenue
- **Optimal_Price** (Target Variable)



Input Features

These 10 input features are used by the ML models to predict the optimal price.



Current_Price

The existing price of the product.



Demand & Demand_Index

Market demand signals and indexed demand score.



Customer_Segment

Target customer category and profile.



Promotion & Discount_Percent

Active promotional campaigns and discount levels.



Competitor_Price

Competitor pricing for comparable products.



Inventory

Current stock levels affecting pricing urgency.



Conversion_Rate

Rate at which views convert to purchases.



Revenue

Historical revenue data for context.

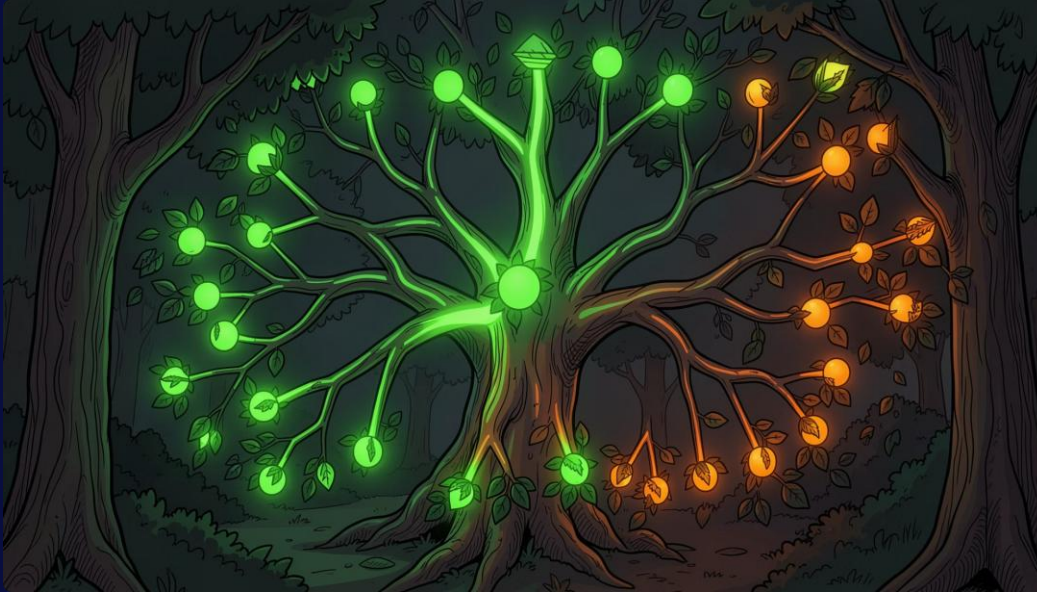


Target Variable

Optimal_Price

The ML models are trained to predict the **Optimal_Price** — the price point that maximizes revenue given current market conditions, demand, inventory, and competitor activity.

Model 1: Random Forest Regression



Purpose

- Predict optimal product prices
- Analyze the impact of demand, inventory, and competitor pricing
- Identify important features affecting pricing decisions

Advantages

High Accuracy

Robust predictions across varied market conditions.

Non-Linear Relationships

Handles complex pricing dynamics naturally.

Feature Importance

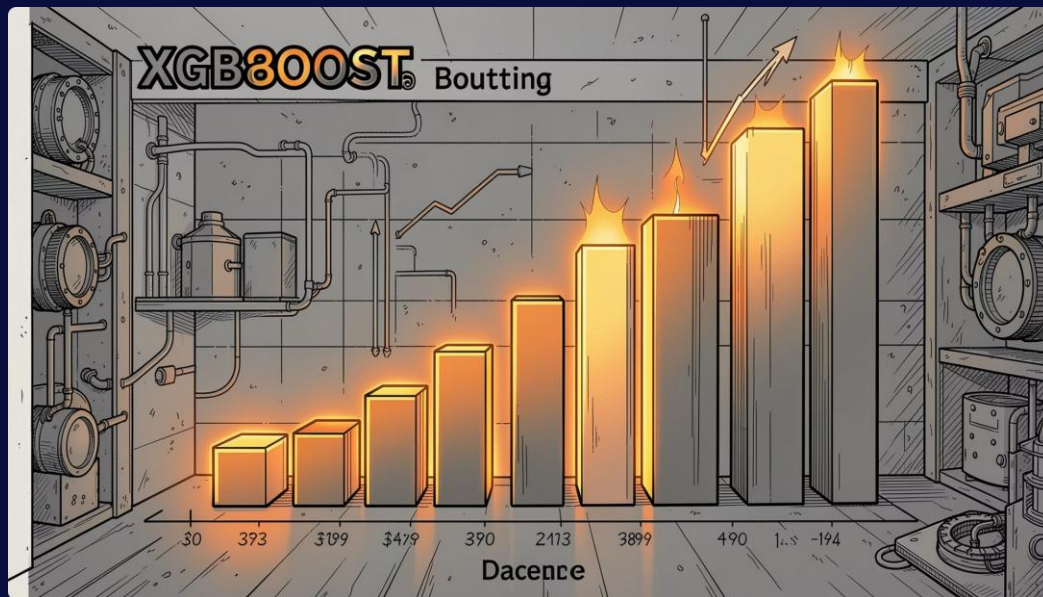
Explains which factors drive pricing decisions.

Model 2: XGBoost Regression

Purpose

- Generate accurate dynamic price recommendations
- Improve revenue optimization and pricing performance

Advantages



High Prediction Accuracy

Gradient boosting delivers state-of-the-art results.

Fast & Efficient

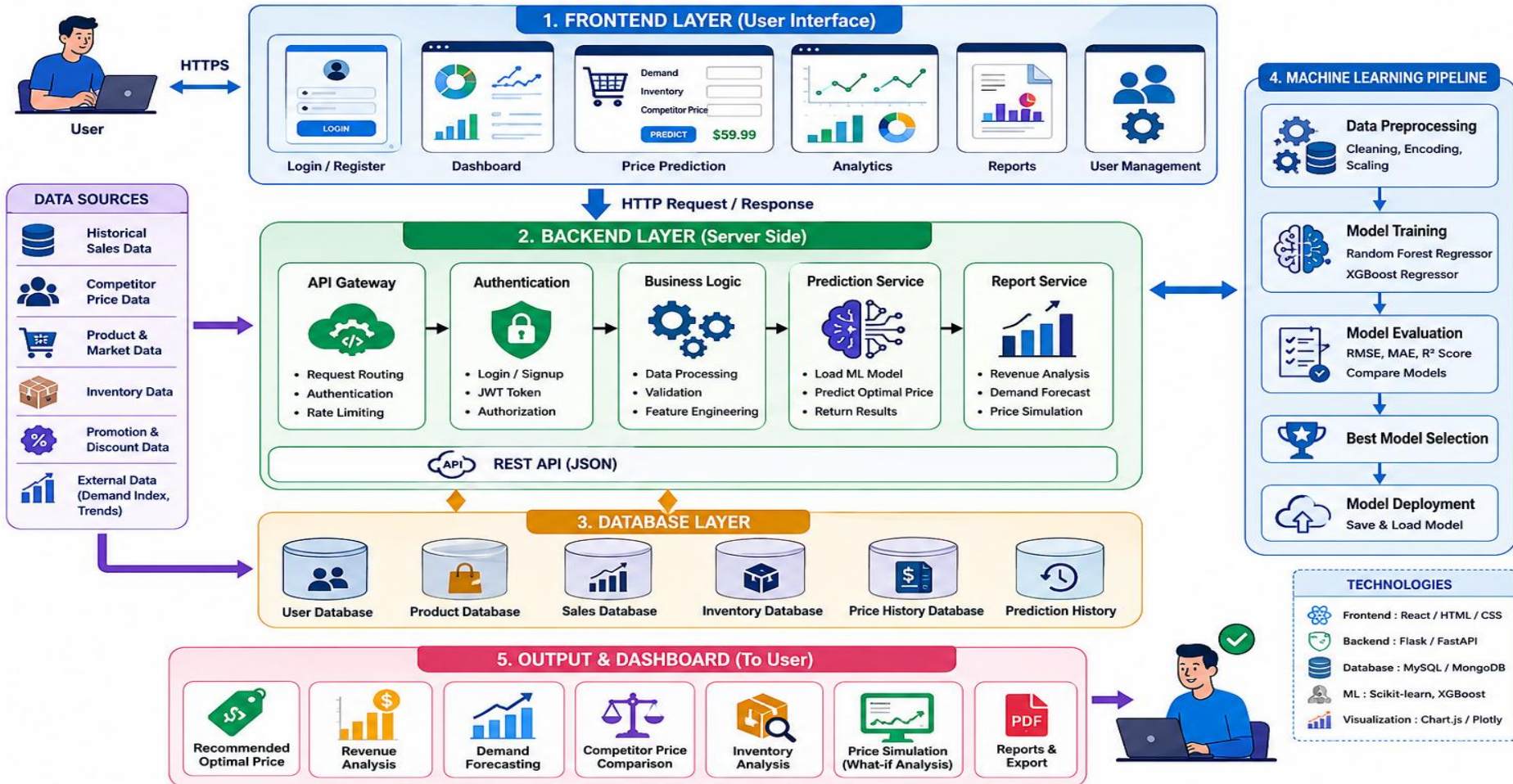
Optimized for speed in production environments.

Scalable

Suitable for large datasets with many features.

DYNAMIC PRICING ENGINE USING MACHINE LEARNING

SYSTEM ARCHITECTURE (FRONTEND + BACKEND)



NAVIGATION

Dashboard

Dynamic Pricing

Demand Forecast

Customers

Competitor

Audit Log

AI Dynamic Pricing Engine

Real-time market tracking, intelligence ingestion, and machine learning price recommendation engine.

CURRENT PRICE

₹1,200

Market Equilibriance

RECOMMENDED PRICE

₹817

-31.9% optimal change

REVENUE IMPACT

₹14,029

+14.2% rev. growth

CONFIDENCE SCORE

96%

99.8% reliability margin

Live Price Recommendation Center

ACTIVE INGESTION

Current Price

₹1,200

Baseline

Recommended Price

₹817

-31.9%

PRICE INCREASE

31.9%

Optimal Markup

AI CONFIDENCE

96%

Very High Accuracy

Revenue Impact Simulator

Price Adjustment Percent

+5.5%

-30%

0% (Base)

+100%

EXPECTED REVENUE

₹15,28,062

EXPECTED DEMAND

1,207

PROFIT CHANGE

+₹37,542

Demand Heatmap



Less More Legend

Feature Importance

Demand

45%

AI Insights Feed

AI PREDICTOR

Pricing Timeline

Price Optimized



NAVIGATION

- Dashboard
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EXPECTED REVENUE

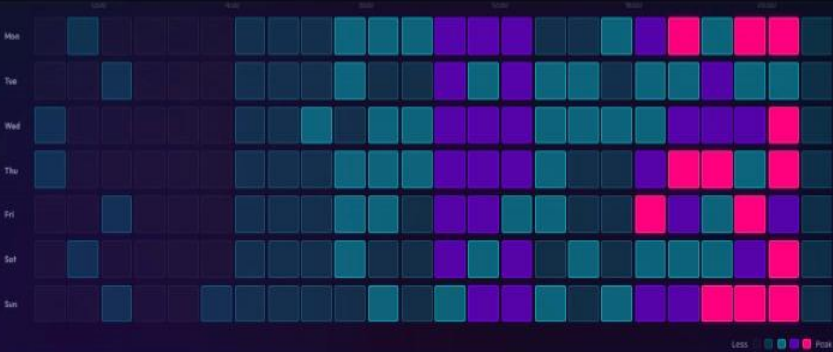
₹15,28,062

EXPECTED DEMAND

1,207

PROFIT CHANGE

+₹37,542



Feature Importance



AI Insights Feed

- AI PREDICTOR**
Demand surge expected at 7 PM based on local traffic forecasting.
2 mins ago
- AI MONITOR**
Inventory running low. Exhaustion predicted in 14 hours.
5 mins ago
- COMPETITOR WATCH**
Flipkart dropped category prices by 8% — review recommended.
15 mins ago
- REVENUE STRATEGY**
Revenue opportunity: upward price elasticity holds for next 4 hours.
20 mins ago

Pricing Timeline

- Price Optimized**
AI optimized pricing to ₹1,350 (+12.5% increase).
10:00 AM
- Demand Spike**
Real-time traffic spiked by 28% on matched inventory.
10:15 AM
- Inventory Alert**
Available inventory dipped below safety margin (45 units).
10:40 AM
- Competitor Action**
Reliance Digital reduced pricing on equivalents by 3.5%.
10:50 AM

Smart Key Performance Indicators

REVENUE ₹2,45,230



DEMAND 78%



ORDERS 1,324



CONV. RATE 3.42%



INVENTORY 91%



NAVIGATION

Dashboard

Dynamic Pricing

Demand Forecast

Customers

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Dynamic Pricing

Predict the optimal price using ML models based on real-time market signals

Price Prediction Inputs

XGBoost

ML Model

CURRENT PRICE (₹)

1200

COMPETITOR PRICE (₹)

1280

DEMAND (UNITS)

200

INVENTORY (UNITS)

50

CONVERSION RATE (0-1)

0.12

DEMAND INDEX (0-1)

0.75

PROMOTION ACTIVE

Yes

DISCOUNT %

0

Predict Optimal Price

Explain Prediction

Prediction Result

XGBoost

Recommended Price

₹816.9

Price Change

-31.9%

Confidence Score

96.0%

Revenue Impact

₹-14,029

Model Used

XGBoost

- NAVIGATION
- Dashboard

Dynamic Pricing

Demand Forecast

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Audit Log

Dynamic Pricing

Predict the optimal price using ML models based on real-time market signals

Price Prediction InputsXGBoostML Model

CURRENT PRICE (₹)

1200

COMPETITOR PRICE (₹)

1280

DEMAND (UNITS)

200

INVENTORY (UNITS)

50

CONVERSION RATE (0-1)

0.12

DEMAND INDEX (0-1)

0.75

PROMOTION ACTIVE

Yes

DISCOUNT %

0

Predict Optimal Price

Explain Prediction

Prediction ResultXGBoost

Recommended Price₹816.9

Price Change-31.9%

Confidence Score96.0%

Revenue Impact—

Model UsedXGBoost

AI Explanation

Price decreased by 31.9% primarily due to current_price. High demand supports premium pricing. Confidence in this recommendation: 96.0%

Current price: \$1200.00

Medium inventory (50 units)

NAVIGATION

- Dashboard
- Dynamic Pricing
- Demand Forecast**
- Customers
- Competitor
- Audit Log

Demand Forecast

AI-powered demand prediction using time-series analysis with seasonal patterns

Forecast Settings

PRODUCT ID

PROD-001

DAYS AHEAD

30

SEASONALITY

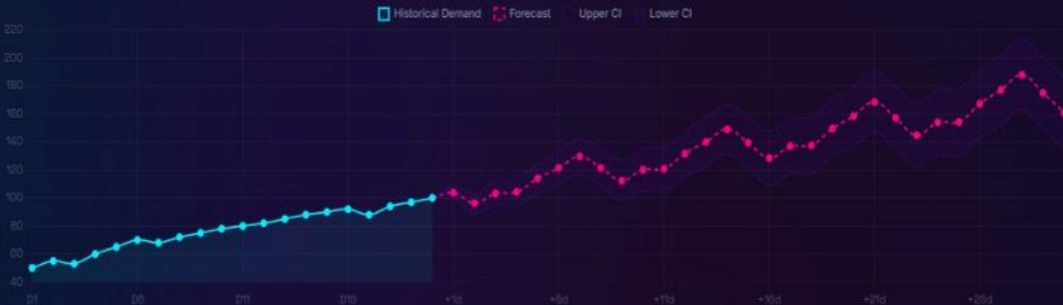
Weekly

HISTORICAL DATA (COMMA-SEPARATED)

50,55,53,80,85,70,88,72,75,78,80,82,85,88,90,92,88,94,97,100

Run Forecast

Demand Forecast Chart



Forecast Insights

Avg Forecasted Demand	139 units/day
Trend Direction	HIGH
Prediction Accuracy	96.8%
CI Lower Range	89.5 - 162.6
CI Upper Range	103.1 - 213.3

Inventory Recommendation

Recommended Reorder Point	655 units
Recommended Order Quantity	1311 units
Days of Supply	1.9 days
Stock Status	Understocked



NAVIGATION

- Dashboard
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Customer Segments

AI-driven customer classification with personalized pricing recommendations

Segment Distribution



Classify a Customer

PURCHASE FREQUENCY

12

AVG ORDER VALUE (€)

1200

CUSTOMER LIFETIME VALUE (€)

14400

LOYALTY SCORE (0-1)

0.85

DAYS SINCE LAST PURCHASE

5

Classify Customer

Segment Overview

Live Data

VIP

High-value, loyal customers with high purchase frequency

Discount: 8% • Retention: 95%

Regular

Regular customers with consistent purchase pattern

Discount: 5% • Retention: 75%

Budget Conscious

Price-sensitive customers who respond to discounts

Discount: 15% • Retention: 60%

Inactive

Customers with low activity, need re-engagement

Discount: 20% • Retention: 40%

New

New customers, acquisition phase

Discount: 10% • Retention: 50%

Classification Result

VIP

Customer Segment

VIP

Segment Description

High-value, loyal customers with high purchase frequency

Recommended Discount

8%

Retention Probability

95%

Price Elasticity

—

Churn Risk

—

Pricing Strategy

[object Object]



NAVIGATION

- Dashboard
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Competitor Analysis

Real-time competitive pricing intelligence and market positioning

Analyze Competitors

PRODUCT ID

PROD-001

YOUR CURRENT PRICE (₹)

1200

Competitor Prices (₹)

Amazon

1280

Flipkart

1300

Meesho

1180

Reliance Digital

1300

+ Add Competitor

Analyze

Market Intelligence

4 Competitors

Amazon

₹1,280

+8.3%

Flipkart

₹1,320

+10.0%

Meesho

₹1,180

-1.7%

Reliance Digital

₹1,300

+8.3%

Avg Competitor Price

₹1,270

Your Price Position

-

Price Difference

-5.8%

Recommended Action

Your price is competitive and offers good value

Market Share Potential

25.0%

Price Comparison



Pricing Strategy

Strategy

Opportunity to increase price to ₹1235.00

Suggested Price

₹1,235

Based on 4 competitors. Market avg ₹1270, optimal price ₹1235.



NAVIGATION

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Audit Log

Audit Log

Complete trail of all pricing decisions, forecasts, and system events

PRICING EVENTS

3

Total predictions made

FORECAST EVENTS

2

Demand forecasts run

COMPETITOR SCANS

2

Market analyses

CUSTOMER EVENTS

2

Segmentations run

Activity Feed

Refresh

Search actions, prices, reasons...

All Events

TIMESTAMP	EVENT TYPE	RECOMMENDED PRICE	CONFIDENCE	REASON / NOTES	STATUS
2026-06-10 10:30	Pricing Predict	₹1,350	94%	Demand high, competitor prices above baseline	✓ OK
2026-06-10 10:15	Competitor Scan	—	—	Market scan triggered by demand spike alert	✓ OK
2026-06-10 09:55	Forecast Update	—	87%	Weekly seasonality pattern applied — 30d ahead	✓ OK
2026-06-10 09:45	Customer Segment	—	—	VIP customer classified — loyalty score 0.92	✓ OK
2026-06-09 16:30	Pricing Predict	₹1,280	91%	Competitor Meesho dropped price by 3%, adjusted accordingly	✓ OK
2026-06-09 14:20	Competitor Scan	—	—	Scheduled daily market intelligence scan	✓ OK
2026-06-09 12:05	Forecast Update	—	89%	Monthly seasonality re-calculation with 20 data points	✓ OK
2026-06-08 18:10	Pricing Predict	₹1,200	78%	Low demand period — baseline price maintained	⚠ Warn
2026-06-08 08:00	System	—	—	AI Engine restarted — models reloaded successfully	✓ OK
2026-06-07 16:02	Customer Segment	—	—	Budget_Conscious segment — 15% discount recommended	✓ OK

NAVIGATION

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Audit Log

Complete trail of all pricing decisions, forecasts, and system events

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PriceIQ Assistant

Online - ML Models Loaded

Price Predict Forecast Competitors

Intelligence. You can ask me about:

- Dynamic pricing strategies
- Demand forecasting
- Customer segmentation
- Competitor analysis
- ML model explanations

Type [help](#) to see all available commands.

09:08 PM

Ask about pricing, demand, competitors...

PriceIQ uses live backend data - Responses may vary

Key Features

These seven features work together to support smarter, faster, and more adaptive pricing decisions.

Live Price Recommendation

Generates dynamic pricing suggestions in real time based on current conditions.

Demand Forecasting

Predicts future customer demand to help guide better pricing and planning.

Competitor Price Tracking

Monitors competitor pricing to keep recommendations aligned with the market.

Revenue Impact Simulation

Estimates how pricing changes may affect revenue before deployment.

Inventory-Based Pricing

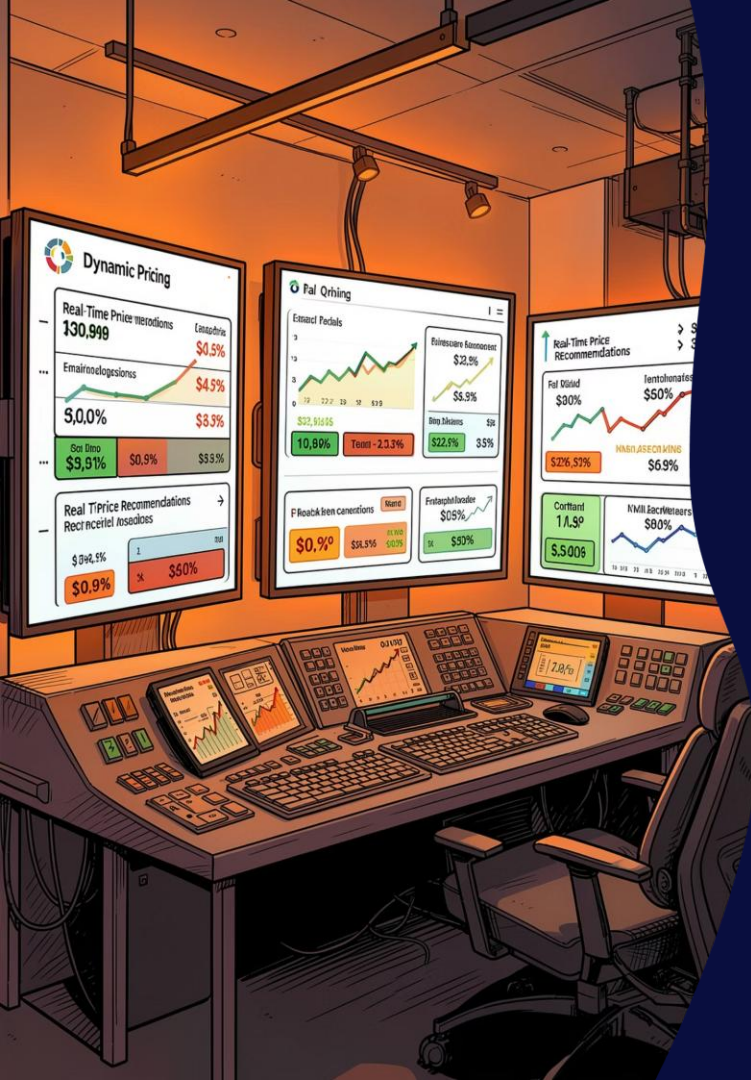
Adjusts pricing based on stock levels to support inventory efficiency.

Explainable AI (Feature Importance)

Shows which factors influence pricing decisions to improve transparency.

AI Chatbot

Provides conversational support for exploring pricing insights and actions.



Future Outcomes

01

Real-Time Price Recommendations

Generate dynamic prices based on live demand, inventory, and competitor data.

02

Demand Forecasting & Revenue Analysis

Support forecasting and track revenue impact before implementing changes.

03

Competitive Pricing Strategies

Track competitor prices and suggest strategies to stay competitive.

04

Explainable Decisions

Provide transparent pricing rationale using feature importance.

Thank,
you!